



A Simple Guide to Building Your Own Linux Kernel

This article is for all the newbies who want to learn how to build their own Linux kernel. Following the steps mentioned will demonstrate how easy the task is!

Please note that the steps mentioned were done on Ubuntu 10.04, but would remain the same for any Linux distribution.

Pre-requisites

Git is the utility for version control on Linux. Install it with a simple `sudo apt-get install git-core`. You also need the curses library development files—install the package with `sudo apt-get install libncurses5-dev`.

Next, verify the current Linux kernel version with the `uname -a` command so you can download the relevant kernel source to build using the `git clone` command. I've used Linux 2.6 in this example, as you can see in Figure 1.

Configuring the kernel

Now that you have the source, you need to configure it. You can use the command `make menuconfig` if you are experienced on what configuration parameters to set. If you are new to this, I would suggest you use the default configuration—copy the existing config file to the kernel source directory with, for example, `cp /boot/config-2.6.32-38-generic .config`

Now, run `make oldconfig` to start the configure process. Please note that here you will be prompted to answer 'Yes' or 'No'. If you are unsure, just keep on hitting 'Enter' (that is a default 'Yes').

Naming the kernel

Next, it might be good to give your kernel a name. For this, open the *Makefile*, and edit the lines below:

```
VERSION = 2
PATCHLEVEL = 6
SUBLEVEL = 32
EXTRAVERSION = -dips
NAME = Building My Kernel
```

Now, issue the `make` command. This should take a couple of hours to complete.

Installing the kernel

Now, issue the following command to install all the kernel modules: